

Summer Intern Quantitative Finance AI Project

Objectives:

1. Develop an AI-based automation framework to generate credit ratings for public companies using market data
2. The underlying finance theory is KMV's Point-In-Time approach and Time-Consistent Through-The-Cycle Credit rating approach
3. Deploy AI programs that consist two main components: one component is for accessing data and information, another for credit assessment

Project Overview:

Credit ratings are a cornerstone of bank risk management. However, methodologies for generating ratings vary significantly depending on company type and data availability. In addition, acquiring the necessary data and effectively applying these methodologies present practical challenges. However, recent development in AI provides great tools for credit ratings of public-traded companies.

This project requires a team with strong quantitative expertise and solid technical capabilities in data engineering and software development to design, implement, and maintain a robust and scalable rating framework. Below is a breakdown of the project.

- Introduction to Credit Risk Management, as well as LLM/AI
 - Credit Rating: PIT, TTC and TiC
 - LLM, AI and Claude Code/Codex
- Introduction to Financial Market
 - Stock, Bond and Options
 - Option Pricing
 - Balance Sheet & other market information
 - Debt and Liabilities, Market Cap, Dividends, Asset return, Risk-free rate
- Data and Automation
 - AI Agent to access market data
 - Programming language: Python and Javascript
 - Data file format: JSON, CSV, EXCEL
 - Data cleansing and preparation
- KMV/TiC/Agency/Bank Credit Ratings
 - Data
 - Time Series
 - Functions and EM algorithm
- AI Implementation Automation
- Rating Conversion: PIT to TTC